Ant Colony Optimization: A Short Survey

Ants find short paths between nest and food without any central control. Each ant deposits pheromone as it walks. Shorter paths are completed faster, so they accumulate pheromone more quickly, which attracts more ants: a positive feedback loop. This indirect coordination through the environment is called stigmergy.

Pheromone evaporation is essential. Without it, the colony would lock in on the first path found, even a poor one. Evaporation acts as forgetting: trails that are not reinforced fade, letting the colony adapt when conditions change.

Applications

Dorigo's Ant System applied these ideas to the travelling salesman problem. Artificial ants build tours probabilistically, biased by pheromone and distance; good tours deposit more pheromone. Variants are used for vehicle routing, network packet routing and job-shop scheduling.

Comparison with other collective decision systems

Like honeybee nest-site selection, ant foraging is decentralized: no individual has global knowledge, yet the group converges on good solutions through local signals, recruitment and thresholds. The balance between exploitation (follow strong trails) and exploration (random wandering) mirrors many learning algorithms.